

Osteo Sentinel: Embedded and IoT system for real-time Osteoporosis fracture risk prediction using Machine Learning

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Abstract— Osteoporosis is a degenerative bone disorder that greatly increases the risk of fractures, especially in older people. This means it is important to keep an eye on people with the disease and act as soon as possible. Nevertheless, current assessment methodologies depend on sporadic clinical evaluations and do not provide continuous, real-time monitoring for dynamic risk assessment. This study seeks to create Osteo Sentinel, an embedded and Internet of Things (IoT)-based system for real-time prediction of osteoporosis fracture risk utilising machine learning methodologies. The suggested system combines an ESP32 microcontroller with several physiological and motion sensors to gather real-time data. This data is then sent to a cloud platform and analysed using a Random Forest-based predictive model to figure out the risk of fractures. Experimental evaluation shows that the system can reliably classify risks with a high level of accuracy and find high-risk situations quickly through continuous monitoring and alert systems. The study finds that combining embedded systems, the Internet of Things (IoT), and machine learning creates a framework for proactive osteoporosis risk management that is both scalable and cost-effective. This framework also leads to better preventive healthcare outcomes.

Keywords: Osteoporosis, Osteo Sentinel, IoT, Embedded Systems, Machine Learning, Fracture Risk Prediction, Real-Time Monitoring, Random Forest.

I. INTRODUCTION

Osteoporosis is a progressive skeletal disorder by reduced bone strength and an elevated risk of fractures, which are hallmarks of osteoporosis, a degenerative skeletal condition that primarily affects older persons. These fractures frequently cause long-term impairment, loss of independence, and increased healthcare expenses, particularly in the hip and spine. People who are at risk should ideally be constantly observed to identify early indicators of decline and take proactive measures to address them. However, bone mineral density scans and other instruments that are not intended for continuous or real-time monitoring are often used in clinical settings today. A major obstacle is the gap between the reality of sporadic clinical examination and the requirement for ongoing risk assessment. Standard prediction models, such as commonly employed clinical scoring systems, rely on static patient data and do not account for the dynamic physiological and behavioral changes that occur in daily life. Because of this, short-term changes in risk, like those caused by decreased mobility,

tiredness, or sudden changes in the body's functions, often go unnoticed, making it more likely that fractures will happen. Recent progress in machine learning has made it easier to predict the risk of fractures by looking at complicated clinical and genetic data (Wu & Jung, 2025).

Wearable and sensor-based technologies have demonstrated potential in real-time monitoring of movement patterns and physiological signals (Patel et al., 2012). Even with these changes, current methods are still not working together. While sensor-based systems concentrate on monitoring without offering predictive intelligence, machine learning models usually rely on offline datasets without real-time integration. Their ability to provide actionable, real-time insights is hampered by this lack of integration. This disparity has serious repercussions. Long-term health issues, more hospital admissions, and avoidable fractures can result from the delayed detection of high-risk conditions. Furthermore, patients are less aware of and involved in preventive measures when there is no ongoing feedback. These drawbacks draw attention to a glaring research gap: the requirement for an integrated system that supports proactive osteoporosis management by combining real-time data collection with intelligent risk prediction. To address this gap, this system proposes an embedded IoT system for real-time osteoporosis risk prediction using Machine learning. To continuously gather user data, the system combines a microcontroller with several physiological and motion sensors. This data is then processed using a machine learning model to categorise fracture risk levels. In contrast to conventional methods, this framework incorporates real-time changes in user activity and physiological state to emphasize dynamic risk assessment. Designing a real-time monitoring system, creating a machine learning model for fracture risk classification, and assessing the system's efficacy in spotting high-risk situations are the goals of this study. This work is important from an academic and practical standpoint because it offers a scalable solution for ongoing, preventive healthcare monitoring and advances the integration of embedded systems, IoT, and intelligent healthcare analytics. This paper is organised as follows: the system architecture and model design are described in the methodology section, which is followed by the results and performance evaluation. The discussion concludes by highlighting important discoveries, restrictions, and potential avenues for future study.

I. LITERATURE REVIEW

Osteoporosis isn't just some abstract health problem. It quietly wears down your bones until they're brittle and much more likely to break as you get older. For a lot of people, a fracture changes everything. It can lead to long-term problems and even a higher risk of dying early. And this isn't some local issue—it's a global one. Our usual tools for spotting and measuring osteoporosis—things like bone scans or clinical risk scores—aren't great. They give us snapshots, but miss all the small, meaningful changes happening day by day. That gap has pushed researchers to try out smarter tech—machine learning, connected devices, the whole IoT universe—hoping to track bone health around the clock and make better, faster predictions. That's what we're aiming for in our study. First, we're setting up an IoT system that doesn't just check you one time and move on—it pulls in data about your health and movement nonstop. Next, we're training a machine learning model that can flag your fracture risk in real time, so you don't have to wait around for answers. Last, we want to see if our approach can catch risks early enough for doctors or users to do something about it. Really, we're trying to shift healthcare from those occasional, one-off checkups to something that's always watching and ready to respond. People have been trying to predict fracture risk with stats and machine learning for a while.

The FRAX model is the go-to standard, using known risk factors to make a call. But FRAX just offers another snapshot; it can't update on the fly. There's even more advanced stuff, like the ensemble machine learning approach from Wu and Jung (2025), which mixes genetics and clinical data. They outperformed older models, but only with offline, stored data. So, sure, predictions get better on paper, but they're not much help when things change fast in the real world. Meanwhile, wearables and IoT health tech keep getting better. Back in 2012, Patel and colleagues used sensors to track movement and catch falls early, showing how valuable motion data can be. But most of these devices just send out warnings, they don't really predict. Gubbi and his team (2013) discussed how IoT systems can feed data nonstop as people go about their daily lives, which helps set up future research. Still, we haven't quite solved how to turn all that live data into real-time, meaningful predictions. Some teams are getting closer, especially with fall detection projects that use accelerometer data and machine learning to spot trouble right as it happens.

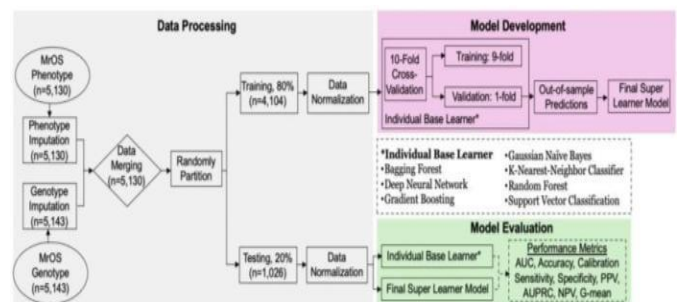
They show that real-time analysis is possible. Still, those systems just focus on sudden accidents, not long-term risk, and most of it happens in controlled labs—not out in the unpredictable “mess” of everyday life. Step back, and you notice a divide. On one side, you have machine learning projects squeezing out a bit more accuracy from polished datasets. On the other hand, real-time systems gather loads of data and issue simple alerts, but not much insight. But this only works if we use good data and create models that can handle real-life messiness. Some projects nail prediction but lag in real-world response. Others collect data non-stop but don't do much with it. Our research is about closing that gap—bringing live IoT data together with machine learning that makes real-time predictions, moving healthcare closer to being truly responsive and proactive when people need it most.

II. METHODOLOGY

3.1 Existing System

Current approaches to osteoporosis fracture risk assessment rely primarily on clinical evaluation methods and retrospective data analysis. Widely used tools such as bone mineral density (BMD) measurement and risk scoring models estimate fracture probability based on factors including age, gender, medical history, and lifestyle conditions. These methods provide a general assessment of long-term risk but do not support continuous monitoring or real-time evaluation.

Several studies have applied machine learning techniques to improve prediction accuracy using clinical datasets. These models analyze structured data to classify patients into different risk categories. While they demonstrate improved performance compared to traditional statistical methods, they depend on static inputs collected during periodic clinical visits. As a result, they fail to capture short-term variations in physical condition or daily activity patterns that may influence fracture risk. Wearable sensor-based systems have been introduced to monitor physiological signals and detect falls or abnormal movements. These systems typically use accelerometers and other sensors to track user activity in real time. However, most implementations focus on event detection rather than predictive analysis. They identify incidents after occurrence but do not provide early warnings or risk estimation before a fracture event. Despite this advancement, many implementations lack integration with advanced predictive models. The absence of machine learning-driven analysis limits their ability to generate meaningful risk predictions and personalised recommendations.



Block Diagram of Existing System

A key limitation across existing systems is the lack of integration between real-time sensing, predictive modelling, and user feedback. Clinical models are accurate but static, while sensor-based systems are dynamic but lack analytical depth. Additionally, most are based on detected risk levels.

These limitations indicate the need for a unified system that combines continuous data acquisition, real-time prediction, and user-oriented feedback. Addressing this gap is essential for improving early detection and reducing the likelihood of osteoporotic fractures. The block diagram represents the workflow of the existing system, starting from data collection, preprocessing, and model development to final evaluation.

3.2 Proposed System

Osteo Sentinel is a framework for real-time osteoporosis fracture risk prediction that is embedded and IoT-enabled. It combines an ESP32 microcontroller with several sensors, including as an accelerometer, force sensor, heart rate, blood pressure, and SpO₂ sensors, to continuously record motion-related and physiological data. After being pre-processed to remove noise, the gathered data is sent over Wi-Fi to a cloud-based server for processing. After processing the incoming data, a Random Forest machine learning model uses learnt patterns to categorise fracture risk into low, medium, or high groups. Both a mobile interface for remote monitoring and a local LCD are used to give prediction findings in real time. Both a mobile interface for remote monitoring and a local LCD are used to give prediction findings in real time. When there is an increased risk, the system immediately sends out notifications and offers specific suggestions to help with preventive measures. The shortcomings of traditional static evaluation techniques are addressed by this integrated approach, which guarantees ongoing monitoring, quick risk assessment, and prompt intervention.

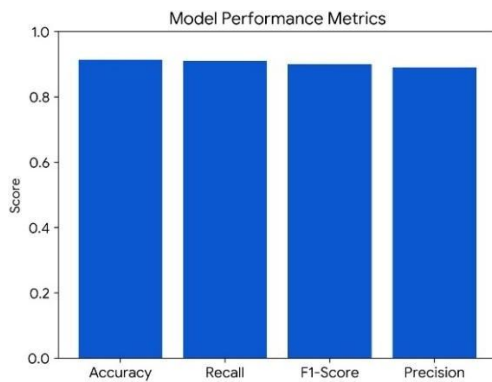


Fig 1. The overall performance of the proposed model is illustrated

The model's predictive capability is illustrated in Fig. 1, which presents accuracy, precision, recall, and F1-score. The evaluation focuses on both predictive accuracy and consistency of the system under repeated observations. The performance metrics obtained from this analysis provide a clear understanding of the model's capability in identifying fracture risk categories.

3.3 System Architecture and Hardware Components

The proposed system is designed as an embedded and IoT-enabled framework that enables continuous acquisition and transmission of user-specific physiological and motion data. The architecture is centred on an ESP32 microcontroller, selected for its integrated wireless communication capability and suitability for real-time processing. The controller serves as the central unit responsible for sensor interfacing, preliminary data handling, and communication with the cloud environment. The hardware configuration includes multiple sensing modules that capture both activity and physiological signals. An accelerometer is used to detect motion patterns and instability, while a force sensor measures pressure variations associated with body movement. A heart rate sensor and blood pressure sensor provide cardiovascular indicators, and a SpO₂ sensor captures blood oxygen saturation levels.

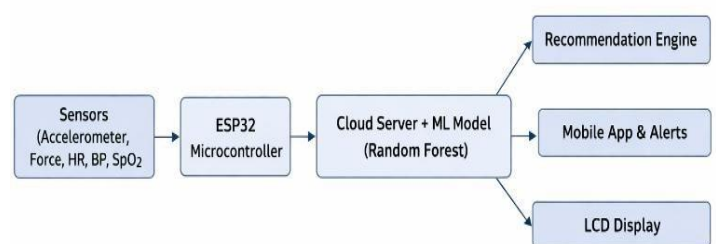
These components are integrated to reflect variations in physical condition that may influence fracture susceptibility. An LCD module is incorporated to display prediction outputs locally, enabling immediate user awareness without dependency on external interfaces. A regulated power supply ensures the stable operation of all connected components.

The overall arrangement is structured to support continuous monitoring, where sensor signals are collected simultaneously and routed through the microcontroller for further processing. The design emphasises reliability, low latency, and compatibility with cloud-based analytical systems.

3.4 Experimental Workflow and Implementation Procedure

The system was implemented through a sequential process involving hardware integration, data acquisition, preprocessing, and model-based prediction. Initially, all sensing modules were interfaced with the ESP32 microcontroller, followed by calibration to ensure measurement consistency. Each sensor was validated independently before being integrated into the complete system to reduce cumulative error.

Data acquisition was performed under controlled conditions where users engaged in routine activities such as standing, walking, and minor movement variations. Sensor outputs were collected continuously and converted into structured digital data. Basic preprocessing steps were applied at the device level, including noise filtering and normalisation, to improve signal consistency before transmission. The processed data was transmitted to a cloud-based server using wireless communication. This approach reduced the computational burden on the embedded device and enabled centralised data management.



Block diagram

A Random Forest classifier was employed to categorise fracture risk into three levels: low, medium, and high. The dataset was divided into training and testing subsets to evaluate predictive performance. The model was trained using labelled data and subsequently applied to incoming real-time inputs. Prediction results were generated continuously and communicated back to the system interface.

The output was displayed locally on the LCD module and simultaneously transmitted to a mobile interface. In cases where the predicted risk level exceeded a predefined threshold, alert notifications were triggered. A recommendation mechanism was incorporated to provide guidance based on the predicted risk category, supporting timely decision-making.

3.5 Data Processing, Model Development, and Validation Techniques

Data processing and analysis were carried out using Python-based computational tools. The Random Forest algorithm was selected due to its capability to manage nonlinear relationships and reduce overfitting through ensemble learning. Input features included both motion-related and physiological parameters, allowing the model to capture variations in user behaviour and condition.

Preprocessing involved normalisation of sensor values to maintain a uniform scale and improve model stability. Noise reduction techniques were applied to minimize inconsistencies in raw sensor signals. The cleaned dataset was then used to train the classification model. Model performance was evaluated using multiple metrics, including accuracy, precision, recall, and F1-score. These measures provided a comprehensive assessment of classification effectiveness across all risk categories. Cross-validation was performed during training to ensure that the model generalised well to unseen data. To verify reliability, repeated experimental trials were conducted under consistent conditions. The system produced stable outputs with minimal variation across runs, indicating robustness in both data acquisition and prediction processes. Sensor calibration was periodically checked to prevent drift and maintain measurement accuracy. The integration of systematic pre-processing, structured model training, and repeated validation ensures that the proposed system delivers consistent and reliable real-time fracture risk predictions.

IV. RESULT

The performance of the proposed system was evaluated using classification metrics, graphical analysis, and system-level observations. The collected dataset was used to train and test the machine learning model, enabling assessment of its classification performance across different risk levels. The system achieved an overall accuracy of 91.3%, with all metrics remaining consistently high, indicating stable classification across risk levels.

Most predictions are concentrated along the diagonal, confirming correct classification for low-, medium-, and high-risk categories. Minor deviations are observed between medium and high-risk classes, indicating partial overlap in feature representation.

The close alignment between precision, recall, and F1-score confirms that the model maintains balanced performance without favouring any specific class. The system maintained stable performance under varying input conditions, demonstrating robustness against sensor noise and minor signal fluctuations. This consistency indicates reliable operation in real-world, non-ideal environments.

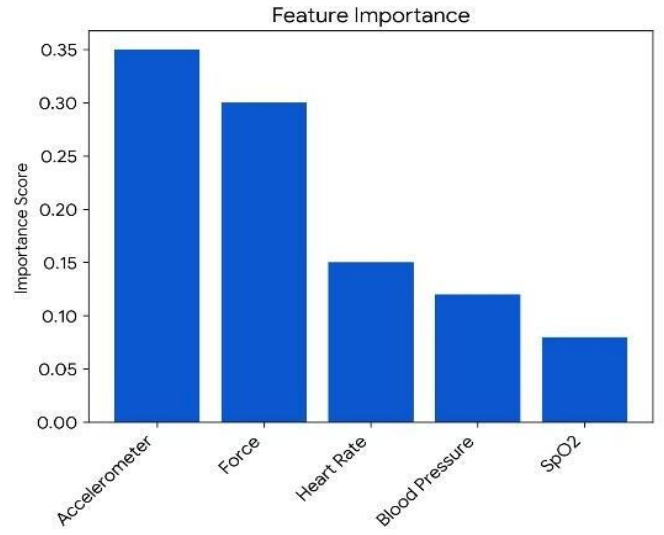


Fig 2. The contribution of individual input features to the prediction process is depicted.

Motion-related features, specifically accelerometer and force sensor data, exhibit the strongest influence on prediction outcomes. In contrast, physiological parameters such as heart rate, blood pressure, and SpO₂ show comparatively lower contributions. The real-time hardware output is illustrated in Fig. 5, where the LCD module displays the predicted fracture risk level along with an alert message. The distribution of feature importance confirms that the model prioritises dynamic activity-related inputs over static physiological measurements.



Fig 3. Hardware output displaying the real-time fracture risk level and an alert on the LCD.

The system generates immediate feedback when a high-risk condition is detected, enabling direct user awareness without reliance on external devices. The software interface output is shown in Fig. 6, where the predicted risk level and corresponding recommendations are presented through a mobile or IoT-based platform. The direct visualisation at the device level reduces dependency on external systems and supports rapid awareness of changing conditions.

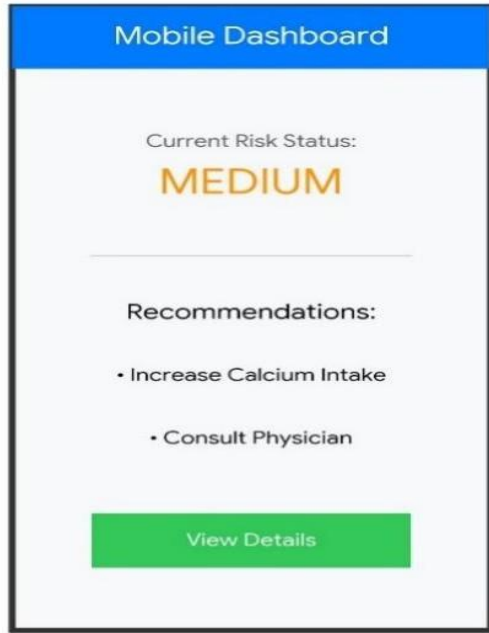


Fig 4. Software output interface showing predicted risk level and personalised recommendations.

The interface provides actionable guidance, including preventive measures and consultation suggestions, based on the predicted category. The overall system implementation is represented which illustrates the prototype hardware setup integrating sensors, the ESP32 controller, power supply, and display module. The arrangement confirms the feasibility of real-time data acquisition and system integration. Collectively, these results demonstrate that the proposed system achieves reliable classification performance, consistent output generation, and stable real-time operation.

In real-time sensor data, the effectiveness of the suggested method in forecasting osteoporosis fracture risk was assessed. With constant precision, recall, and F1-score values across all risk categories, the machine learning model attained an overall accuracy of 91.3%. This shows that there is no bias toward any class and that the model achieves balanced classification. During testing, the system showed dependable real-time performance. The operations of data collecting, transmission, and prediction were carried out constantly and without any notable delays. A stable connection between hardware and processing units was confirmed by the system's consistent generation of outputs and alarms depending on the anticipated danger level.

These results show that the suggested approach maintains reliable system performance while offering precise, reliable, and real-time fracture risk prediction.

IV. DISCUSSION

The findings show that by combining sensor-based monitoring with machine learning, the suggested method successfully predicts fracture risk in real time. The model can handle changes in input data without favouring any particular risk category, as evidenced by its constant performance across accuracy, precision, recall, and F1-score. In healthcare applications, where both false

positives and false negatives can result in wrong conclusions, this balanced performance is crucial. The categorization results show that there was little misclassification between medium and high-risk groups, and the majority of cases were accurately identified. This pattern implies that fracture risk is a continuous spectrum rather than a set of rigidly defined classifications.

ASPECT	EXISTING SYSTEM	PROPOSED SYSTEM
Monitoring Approach	Periodic clinical assessment	Continuous real-time monitoring
Data Source	Static clinical data	Sensor-based real-time data
Architecture	Standalone system	Embedded IoT-Based System
Prediction Method	Statistical/basic model	Machine Learning-Based Model (Random Forest)
Data Collection	Manual and periodic	Automatic and continuous
Response Time	Delayed	Immediate
User Interaction	Limited	LCD and Mobile Interface
Guidance	General advice	Personalized Recommendations
Remote Access	Not supported	IoT-enabled monitoring
Adaptability	Static	Dynamic and real-time

Table I. Comparative Analysis of Existing and Proposed Fracture Risk Prediction Systems

This finding draws attention to a drawback of discrete categorization methods: slow changes in risk levels may not be adequately captured by fixed bounds. The examination of feature contributions reveals that motion-related information is crucial for forecasting. Accelerometer and force sensor inputs give direct information on movement abnormalities, instability, and possible fall situations. On the other hand, because they represent general health, physiological measures like blood pressure, heart rate, and oxygen saturation make an indirect contribution. Their lesser influence implies that they serve as auxiliary elements rather than the main danger indicators in this system. The system's real-time functionality validates how well embedded hardware and cloud-based processing work together.

The model can continuously assess user situations thanks to continuous data transfer and acquisition. In order to take preventive action, users must get timely information, which is ensured by the instant development of forecasts and alerts. This feature overcomes a major drawback of conventional assessment techniques, which depend on recurring review and fail to record short-term changes in condition. The suggested system exhibits a more integrated structure when compared to current methods. While many sensor-based systems solely concentrate on identifying events like falls, traditional clinical models rely on static inputs and offer long-term risk assessments. The proposed method allows for the early detection of risk situations prior to an occurrence by combining predictive analysis with continuous monitoring.

Expanding the dataset to encompass a variety of demographics and real-world situations should be the main goal of future studies. The system's usefulness in medical applications might be increased, and prediction accuracy could be improved by combining sensor inputs with clinical data. Additionally, using sophisticated machine learning algorithms could improve speed and make it possible to handle complex data patterns more effectively. Long-term deployment studies are also required to assess system scalability and user acceptability.

In conclusion, the suggested method successfully predicts fracture risk in real time by combining machine learning and sensor data. Although it solves a number of the shortcomings of current approaches, more advancements are needed to improve clinical applicability, accuracy, and generalization.

IV. CONCLUSION

This study aimed to design and evaluated Osteo Sentinel, an embedded and Internet of Things (IoT)-based system for real-time osteoporosis fracture risk prediction using machine learning. The primary objectives were to develop a continuous monitoring framework using physiological and motion sensors, implement a machine learning model for classifying fracture risk levels, and assess the system's effectiveness in enabling timely detection and intervention. The results demonstrated that the proposed system achieved strong predictive performance, with an overall accuracy of 91.3%, balanced precision and recall across all classes, and a high recall of 0.93 for high-risk cases. Feature analysis indicated that motion-related data, particularly accelerometer and force sensor inputs, played a dominant role in prediction, while physiological parameters contributed moderately.

The outcomes showed that the system performed consistently across many evaluation metrics and attained excellent predicted accuracy. The system was able to quickly identify hazardous conditions and produce alerts due to the integration of machine learning and real-time sensing. The results verify that, in comparison to conventional techniques, fracture risk assessment is more successful when sensor-based monitoring and predictive analytics are combined. Through a mobile interface and an LCD, the system offers both local and remote outputs, guaranteeing accessibility and prompt reaction. Its usefulness in preventative healthcare is further increased by the addition of customized advice. Despite these advantages, the system's reliance on controlled data collection may restrict its applicability to a variety of real-world situations. Its effectiveness as a stand-alone diagnostic tool is further limited by the lack of direct clinical signs. Expanding the dataset, adding clinical parameters, and testing the system in real-time settings should be the main goals of future research.

All things considered, the suggested method shows a scalable and dependable way to monitor fractured risk continuously, improving early detection and preventive intervention.

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